

function nodes, like:

- a. *Input nodes:*
 - i. HTTP
 - ii. IBM IoT
- b. *Output nodes:*
 - i. HTTP response
 - ii. IBM IoT
 - iii. Twilio (send SMSs using the Twilio service)
- c. *Function nodes:*
 - i. Function
 - ii. Switch

Running Node-RED

Once the Node-RED installation and initial configuration is done, it is ready to be deployed or used. There are three ways in which Node-RED can be run:

- Locally (Docker-container)
- On a device (Raspberry-Pi, Arduino, Android based devices, etc)
- In the identified cloud environment

Where can Node-RED be used?

Node-RED can be used in a range of applications. The major ones are listed below.

- 1) In Bluemix, for connecting to IoT (with ReST and MQTT).
- 2) For binding and connecting to databases (MongoDB).
- 3) For storing IoT data for present and future computation.
- 4) For social media, when taking action and when event driven applications are needed (like Twitter).

Node-RED has over 225,000 package repos, to which it is easy to extend and add new packages. It also has a dedicated

community (*nodered.org*), and is built with robust architecture using Node.js (which is very popular these days).

Node-RED can be used in event driven and fast-to-market applications and services, with easily implementable steps.

Home-OS and Node-RED

There is an alternate approach to visual programming in the IoT space, which uses Home-OS from Microsoft. In this case, users can record macros to activate sequences of actions, such as turning on lights, etc. This feature allows users to set up simple and easy-to-use macros, without the need to learn an abstract visual or verbal representation of the system. More details on Home-OS can be found at <https://www.microsoft.com/en-us/research/project/homeos-enabling-smarter-homes-for-everyone/>. 

References

- [1] <https://mustache.github.io/mustache.5.html>
- [2] <http://nodered.org/docs/getting-started/installation>
- [3] <http://nodered.org/>
- [4] <http://nodered.org/docs/platforms/docker>
- [5] <https://github.com/node-red/node-red/wiki/Architecture-Overview>
- [6] <https://github.com/node-red/node-red>

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